Overview

The **Driver Drowsiness Dataset (DDD)** contains cropped facial images of drivers classified into two categories: **Drowsy** and **Non-Drowsy**. These images have been extracted and cropped from the videos of the Real-Life Drowsiness Dataset.

The frames were captured from the source videos using VLC software. The Viola-Jones algorithm was then applied to detect and extract the face region of interest from each image. This dataset was originally used in the research paper *"Detection and Prediction of Driver Drowsiness for the Prevention of Road Accidents Using Deep Neural Networks Techniques"*.

Dataset Structure

The dataset is organized in a standard image classification folder structure with separate directories for training and testing:

```
dataset/
├── train/
│   ├── drowsy/        ← Images of drowsy faces (for training)
│   └── non_drowsy/    ← Images of alert faces (for training)
└── test/
    ├── drowsy/        ← Images of drowsy faces (held-out test set)
    └── non_drowsy/    ← Images of alert faces (held-out test set)
```

Class Labels

This is a binary classification task with two target classes:
• **Drowsy (1):** Driver appears to be falling asleep or fatigued.
• **Non-Drowsy (0):** Driver is alert and awake.

Data Splits

The dataset comes pre-split into training and test sets.

• **Training:** From the train/ folder — used for model learning.
• **Test:** From the test/ folder — used only for final evaluation.

Image Details

• **Format:** JPEG / PNG
• **Content:** Cropped facial regions of drivers (extracted using the Viola-Jones algorithm).
• **Color:** RGB (3-channel color images).

• **Resizing:** Images should be resized to a uniform size (e.g., 128×128) before feeding to the model.
• **Source:** Frames extracted from the Real-Life Drowsiness Dataset videos using VLC software.

Key Notes for Learners

• The test set must be kept completely separate and used only for final evaluation — never for training or hyperparameter tuning.
• Data augmentation (flipping, brightness changes, zoom, etc.) should be applied.